

Week 2

Keel Bone Chicken Project

05/25/26



College of Engineering

This Week

Weekly Progress

- **Data Collection Updates**
 - New data logger frontend
 - Necessary firmware updates → avoid overheating
 - Corresponding backend updates
- **PCB Updates**
 - Fundamental circuit diagram completed
 - Updated KiCAD full circuit
 - Zoomed in RFID section
 - Finished PCB component for use on new PCB
 - PCB component added to design
- **Database Updates**
 - MongoDB database collection method started
- **Next Week**

Data Collection Updates

New data logger frontend

- Sensors will operate on a dropdown system to save cache memory
- 3D viewer still present to get an idea of the current positioning of the system
- Critical parts of new design stated below:



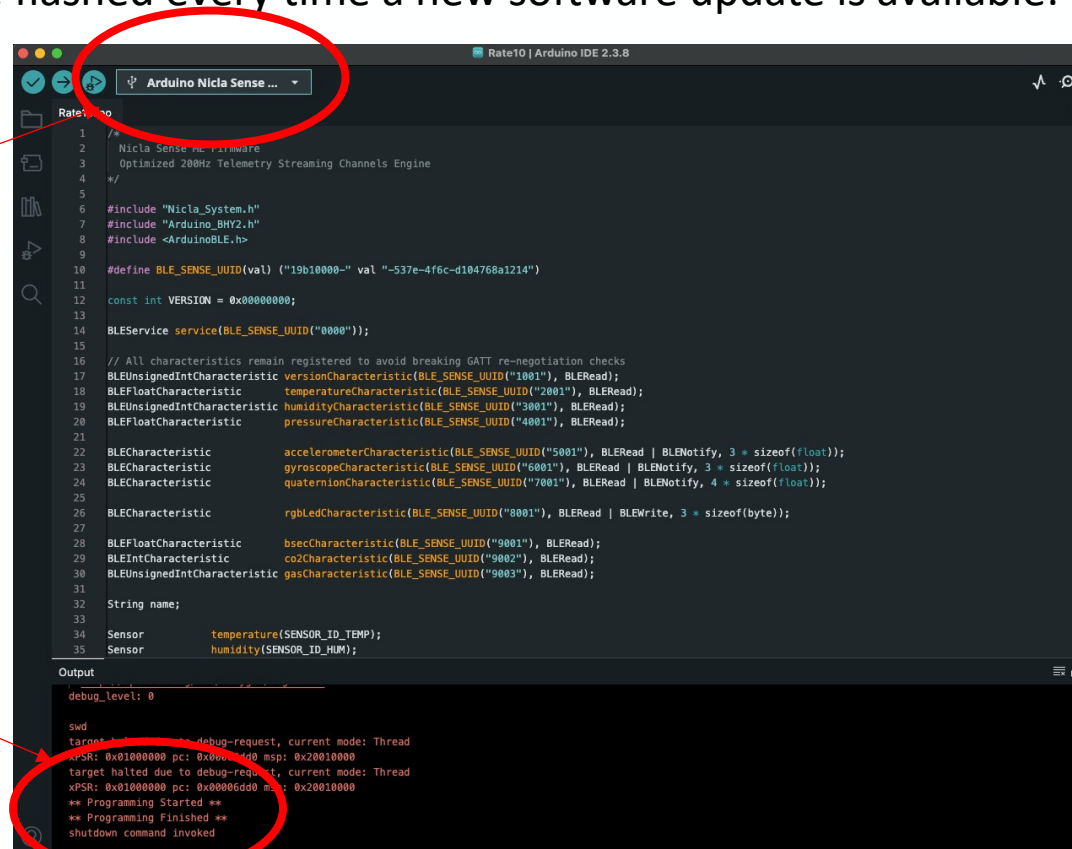
Data Collection Updates

Necessary firmware updates → avoid overheating

- Unnecessary Sensors have been deactivated
- Firmware NEEDS to be flashed every time a new software update is available:

Ensure the correct device is selected

Correct shutdown message will be stated



```
1 // Nicla Sense ME Firmware
2 // Optimized 200Hz Telemetry Streaming Channels Engine
3 //
4
5 #include "Nicla_System.h"
6 #include "Arduino BHY2.h"
7 #include <ArduinoBLE.h>
8
9 #define BLE_SENSE_UUID(val) ("19b10000-" val "-537e-4f6c-d104768a1214")
10
11 const int VERSION = 0x00000000;
12
13 BLEService service(BLE_SENSE_UUID("0000"));
14
15 // All characteristics remain registered to avoid breaking GATT re-negotiation checks
16 BLEUnsignedIntCharacteristic versionCharacteristic(BLE_SENSE_UUID("1001"), BLERead);
17 BLEFloatCharacteristic temperatureCharacteristic(BLE_SENSE_UUID("2001"), BLERead);
18 BLEUnsignedIntCharacteristic humidityCharacteristic(BLE_SENSE_UUID("3001"), BLERead);
19 BLEFloatCharacteristic pressureCharacteristic(BLE_SENSE_UUID("4001"), BLERead);
20
21
22 BLECharacteristic accelerometerCharacteristic(BLE_SENSE_UUID("5001"), BLERead | BLENotify, 3 * sizeof(float));
23 BLECharacteristic gyroscopeCharacteristic(BLE_SENSE_UUID("6001"), BLERead | BLENotify, 3 * sizeof(float));
24 BLECharacteristic quaternionCharacteristic(BLE_SENSE_UUID("7001"), BLERead | BLENotify, 4 * sizeof(float));
25
26 BLECharacteristic rgbLedCharacteristic(BLE_SENSE_UUID("8001"), BLERead | BLEWrite, 3 * sizeof(byte));
27
28 BLEFloatCharacteristic bsecCharacteristic(BLE_SENSE_UUID("9001"), BLERead);
29 BLEIntCharacteristic co2Characteristic(BLE_SENSE_UUID("9002"), BLERead);
30 BLEUnsignedIntCharacteristic gasCharacteristic(BLE_SENSE_UUID("9003"), BLERead);
31
32 String name;
33
34 Sensor temperature(SENSOR_ID_TEMP);
35 Sensor humidity(SENSOR_ID_HUM);
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```

Output

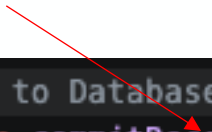
```
debug_level: 0
swd
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x00000000 msp: 0x20010000
target halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x00000000 msp: 0x20010000
** Programming Started **
** Programming Finished **
shutdown command invoked
```

Collection

Corresponding backend updates

- Backend again hosted at: https://nhuang58.github.io/Biomedical_Device_PCB/
- Uses a InterDB Database to store data instead of relying on local storage on a user(s) device
- Storage to the Database based on an API call within the code

```
279 | // Call to Database to record data
280 | function commitRecordToDB(record) {
281 |   if (!db) return;
282 |   const tx = db.transaction(["telemetry_logs"], "readwrite");
283 |   const store = tx.objectStore("telemetry_logs");
284 |   store.add(record);
285 | }
286 |
```

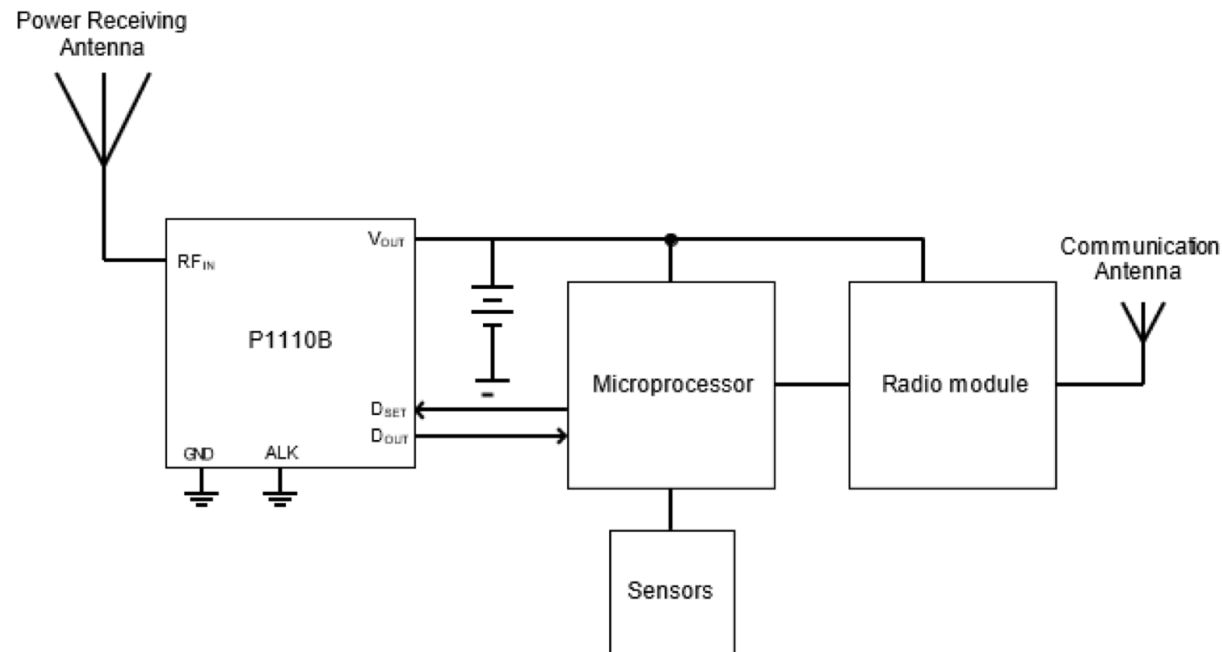


PCB Updates

Fundamental circuit diagram completed

- Based on primitive circuit found in datasheet
- Other circuit specifications needed to be outlined

TYPICAL APPLICATION CIRCUIT

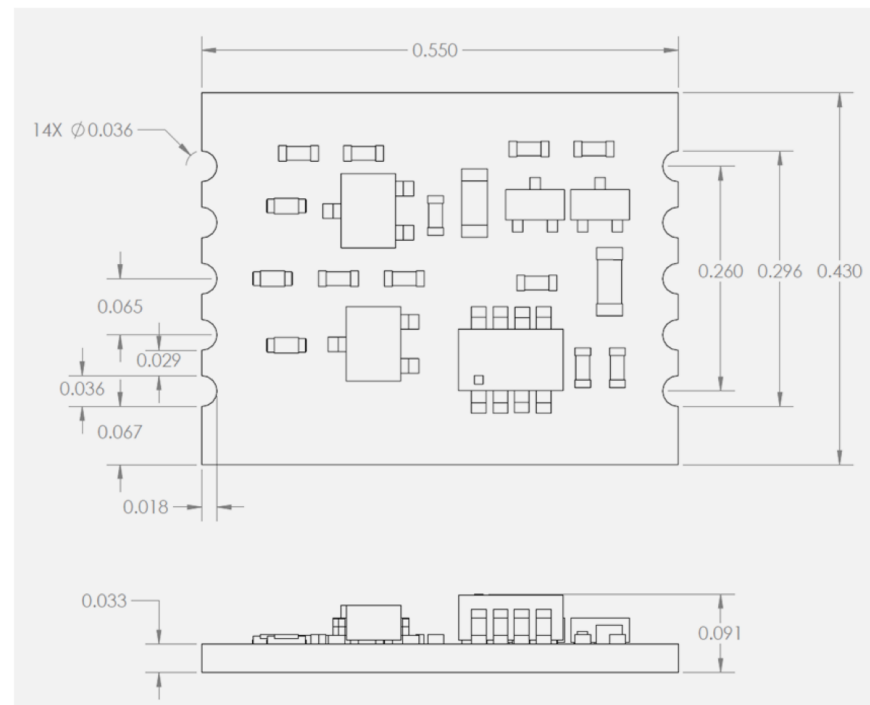


PCB Updates

Fundamental circuit diagram completed

- Mechanical design specifications outlined:
- Units in inches (hard to locate)
- 50 impedance antenna required (The antenna we have is sufficient)
 - $0.550 \times 0.430 \times 0.033$ in³
 - $13.97 \times 10.92 \times 2.31$ mm³
- Sufficient space on board
 - See next slide

MECHANICAL SPECIFICATIONS



Updated KiCAD full circuit



PCB Updates

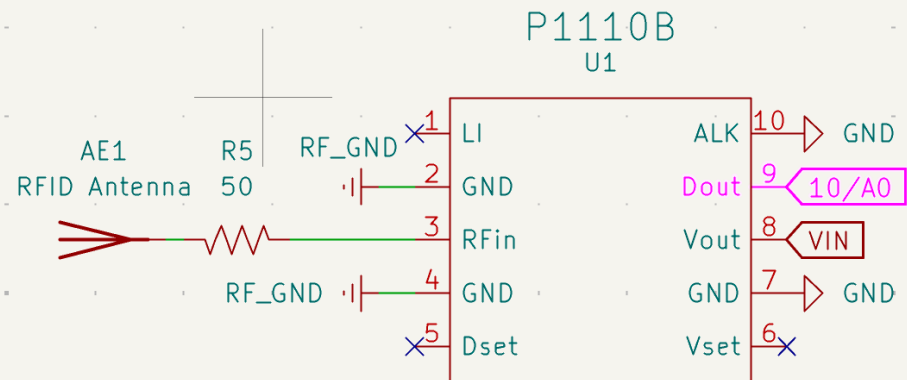
Zoomed in RFID section

- A0 analog input to read data in case of debugging
- VIN connected to battery charger
 - 3.3 V level

Powercast P1110B: RFID → DC Converter

ALK set to DC GND to set voltage level to 3.3 V

10/A0 Pin used to read signal strength level

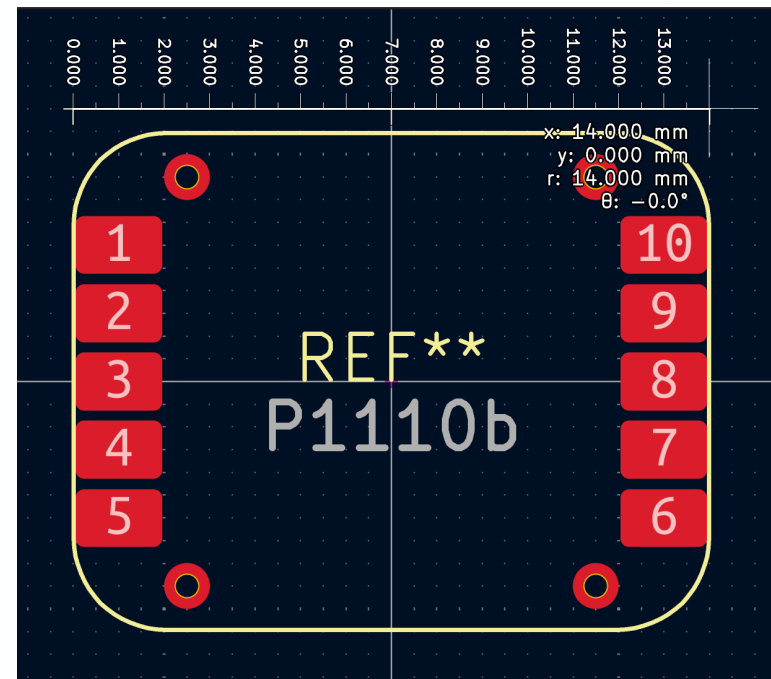
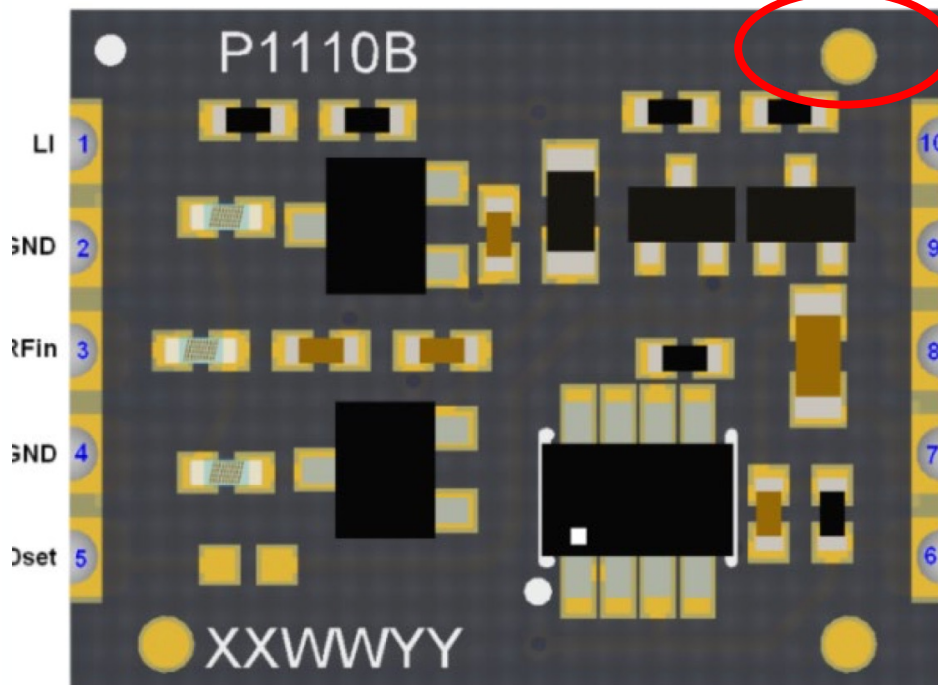


Custom Symbol for P1110b Powercast Harvester

PCB Updates

Finished PCB component for use on new PCB

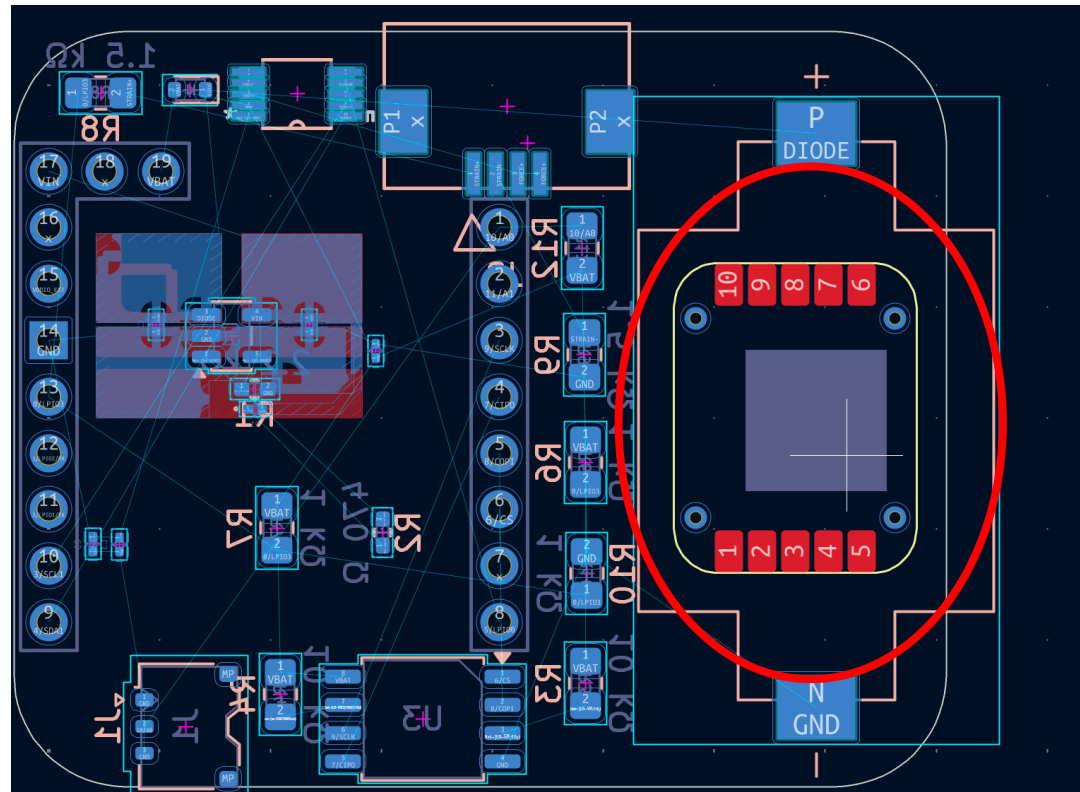
- Allowed correct spacing for the P1110b mounting holes
- 1 mm diameter mounting hole:



PCB Updates

PCB component added to design

- Relative sizing on the back of the PCB seen for reference:
- Not a large component by any regard



Database Updates

MongoDB database collection method started

- Data collection method 3
 - Cloud Storage:
 - Operation: Data would be immediately backed up to the cloud or another service
 - Collection: A remote login would be needed to access data, nothing else
- Potential Solution: MongoDB
 - Runs off AWS servers
 - Free for immediate use (No large-scale data)
 - Perfect for an intermediate cloud solution that is easy to maintain
 - Data can be pulled from the database without having to connect to a Nicla BLE device



Database Updates

MongoDB database collection method started

- Created Cluster (Keel_Bone_Study_Database) for collecting data through MongoDB servers
- Free for now, up to 0.5 GB (.csv files on the order of 0.0001 Gb (0.02%))
- Shouldn't exceed the limits with the size of .csv files collected

☐ **M10** **\$0.08/hour**
Dedicated cluster for development environments and low-traffic applications.

STORAGE	RAM	vCPU
10 GB	2 GB	2 vCPUs

☐ **Flex** **From \$0.011/hour**
Up to \$30/month
For development and testing, with on-demand burst capacity for unpredictable traffic.

STORAGE	RAM	vCPU
5 GB	Shared	Shared

☒ **Free**
For learning and exploring MongoDB in a cloud environment.

STORAGE	RAM	vCPU
512 MB	Shared	Shared

☒ **Free forever!** Your free cluster is ideal for experimenting in a limited sandbox. You can upgrade to a production cluster anytime.

Configurations

Name
You cannot change the name once the cluster is created.

Provider

aws

Google Cloud

Azure

Quick setup

☒ Automate security setup ⓘ

☐ Preload sample dataset ⓘ

Next Week

Next Week's Plan

- Implement MongoDB backend with current web application
 - No need for frontend, simplifies a lot of the work needed
- Write documentation for operation of the device
 - Professor Karcher's request for making sure the device is properly operated
- Showcase Agriculture department how to pull data from the database
 - Introduce MongoDB to Omosalewa and make sure the web application is as intuitive as possible

Thank You

Purdue Polytechnic Institute



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